heorem[section] emma orollary efinition emark ssumption

Causal SCX: Causal Inference

Causal SCX — Multi-Expert Audit of Causal Inference

SCX

2026

Abstract

SCXscx_causal, 3-SCM (1) Disagreement $M \geq 2$ **(1) Causal ConsensusCausal Consensus Bound:** Hoeffding awaiting, M $\{\hat{\tau}_m\}_{m=1}^{M}$ ConsistencyATE τ^* .Proof M , τ^*, **systematic confounding (2)** $\mathcal{Z}_1, \mathcal{Z}_2, \ldots, \mathcal{Z}_M$ M U T Y $\mathcal{Z}_m$, $D(\hat{\tau}_1, \ldots, \hat{\tau}_M)$. **(3)** $\{Z_1, \ldots, Z_K\}$, IV Disagreement Q_k k . **(4) do-calculus** do Disagreement., SCX **Keywords: do-calculus,**

1

——, , awaiting. **SCX**scx_causal(1) Disagreement .Core——**3-SCM** Theorem 1, (1)——SCM $M > 3$ **Causal SCX**——Causal Inference. do-calculus SCX . **:** $T \in \{0,1\}$, $Y \in \mathbb{R}$, $X \in \mathbb{R}^p$.

ATE $\tau^* = \mathbb{E}[Y(1) - Y(0)]$, $Y(t)$ .M $\hat{\tau}_1, \ldots, \hat{\tau}_M$, $\mathcal{M}_m$.

2

2.1

D 1 (SCM)**.** *SCM* $\mathcal{M} = \langle U, V, F, P(U) \rangle$ U, V, $F = \{f_v : v \in V\}$ U . $\mathcal{M}$ $P_{\mathcal{M}}(V)$ $P_{\mathcal{M}}(V \mid \mathrm{do}(t))$.

D 2 ($\mathcal{M}$-awaiting)**.** *SCM* $\mathcal{M}_1, \mathcal{M}_2$ *awaiting,* $\mathcal{M}_1 \equiv_{obs} \mathcal{M}_2$, *:* $P_{\mathcal{M}_1}(V) = P_{\mathcal{M}_2}(V)$.

Theorem 1 (3-SCM): awaiting SCM $\mathcal{M}_1, \mathcal{M}_2, \mathcal{M}_3$, Q ATE:

$$Q(\mathcal{M}_1) \neq Q(\mathcal{M}_2) \neq Q(\mathcal{M}_3) \neq Q(\mathcal{M}_1),$$

Q . M , Disagreement.

2.2 Hoeffding

Hoeffding awaiting: $X_1, \ldots, X_M$, $a_i \leq X_i \leq b_i$, $\varepsilon > 0$,

$$P\left(\left|\frac{1}{M}\sum_{i=1}^{M} X_i - \mathbb{E}\left[\frac{1}{M}\sum_{i=1}^{M} X_i\right]\right| \geq \varepsilon\right) \leq 2\exp\left(-\frac{2M^2\varepsilon^2}{\sum_{i=1}^{M}(b_i - a_i)^2}\right).$$

Core: —— $P(V)$, —— $\hat{\tau}_m$

3 Causal Consensus

3.1

M $\{\hat{\tau}_m\}_{m=1}^{M}$ $\mathcal{Z}_m$, g_m $\mathcal{H}_m$ ATE :

$$\hat{\tau}_m = \hat{\tau}(\mathcal{Z}_m; g_m; \mathcal{H}_m).$$

consensus measure:

$$C_M = 1 - \frac{1}{M(M-1)} \sum_{1 \le i < j \le M} |\hat{\tau}_i - \hat{\tau}_j|.$$

$C_M = 1$; Disagreement $C_M \to 0$.

3.2

$$\hat{\tau}_m - \tau^* = \underbrace{\mathrm{bias}(\mathcal{Z}_m, \mathcal{H}_m)} + \underbrace{\varepsilon_m}.$$

D 3. *consensus offset:*

$$\delta_M = \frac{1}{M} \sum_{m=1}^{M} \mathrm{bias}(\mathcal{Z}_m, \mathcal{H}_m).$$

3.3

T 4 (Causal Consensus — Hoeffding). *(A1)* $\hat{\tau}_m$ $[\tau_{\min}, \tau_{\max}]$*;*

(A2) ε_m $P(V)$ */*

(A3) m_0 $\mathrm{bias}(\mathcal{Z}_{m_0}, \mathcal{H}_{m_0}) = 0$
$\varepsilon > 0$*:*

$$P\left(\left|\frac{1}{M} \sum_{m=1}^{M} \hat{\tau}_m - \tau^* - \delta_M\right| \ge \varepsilon\right) \le 2 \exp\left(-\frac{2M\varepsilon^2}{(\tau_{\max} - \tau_{\min})^2}\right).$$

, $\delta_M = 0$*,* $O(1/\sqrt{M})$ τ^**., ATE,* δ_M τ^* *Scale,* $C_M \to 1$ *.*

Proof Sketch. $\hat{\tau}_m$ $\tau^* + \mathrm{bias}_m + \varepsilon_m$. $\delta_M + \frac{1}{M}\sum_m \varepsilon_m$. (A2) , ε_m $P(V)$, Hoeffding Conclusion. Core: Hoeffding δ_M——Causal Inference"awaiting". □ □

Honest Strike / Honest Critique:
Honest Strike: §7 . **Honest Strike:** , ., 4 , τ^* .

C 5. *4* , $\delta_M \neq 0$ $|\delta_M| > \frac{\tau_{\max} - \tau_{\min}}{\sqrt{2M}}$, $\{\hat{\tau}_m\}_{m=1}^{M}$.

3.4

, M, consensus depth:

$$D(M) = \inf\{d \in \mathbb{N} : \exists d\}.$$

$D(M) > M/2$, —— ..

4

4.1

M $\mathcal{Z}_1, \ldots, \mathcal{Z}_M \subseteq X$. $\mathcal{Z}^*$, : $\mathcal{Z}^*$ T Y . $\mathcal{Z}_m \subsetneq \mathcal{Z}^*$, $\hat{\tau}_m$; $\mathcal{Z}_m \supseteq \mathcal{Z}^*$, .

A 6. *:* $\mathcal{Z}_1 \preceq \mathcal{Z}_2 \iff \mathcal{Z}_1 \subseteq \mathcal{Z}_2$. $\mathcal{Z}^*_{\text{min}}$.

T 7. ***Disagreement****:*

$$\Sigma_{dis} = (|\hat{\tau}_i - \hat{\tau}_j|)_{i,j=1}^{M} \in \mathbb{R}^{M \times M},$$

$\rho(\Sigma_{dis})$. *:*

$$G_m = \frac{\hat{\tau}_m - \hat{\tau}_{m-1}}{|\mathcal{Z}_m| - |\mathcal{Z}_{m-1}|},$$

$|\mathcal{Z}_m|$ $|\mathcal{Z}_1| < |\mathcal{Z}_2| < \cdots$. $\mathcal{H}_0$*:*

(i) $\rho(\Sigma_{dis}) \xrightarrow{p} 0$ *as* $n \to \infty$*;*

(ii) $\sup_m |G_m| \xrightarrow{p} 0$ *as* $n \to \infty$*;*

(iii) $\hat{\tau}_m$ *:* $\exists m_0 : \forall m \geq m_0, |\hat{\tau}_m - \tau^*| < |\hat{\tau}_{m-1} - \tau^*|$.

(i) (ii) .

Proof Sketch. $\mathcal{Z}_m$ $\mathcal{Z}_m \supseteq \mathcal{Z}^*_{\text{min}}$, $\hat{\tau}_m$ τ^*: $\hat{\tau}_m \xrightarrow{p} \tau^*, \forall m$. $\Sigma_{\text{dis}} \xrightarrow{p} 0$ $G_m \xrightarrow{p} 0$. , $\mathcal{Z}_m \not\supseteq \mathcal{Z}^*_{\text{min}}$, , $\hat{\tau}_m \xrightarrow{p} \tau^* + b_m$ $b_m \neq 0$. , b_m Σ_{dis} . , Appendix. □ □

D 8.

$$D_{conf} = \frac{\rho(\Sigma_{dis})}{\frac{1}{M}\sum_{m=1}^{M} \widehat{\mathbb{V}\text{ar}}(\hat{\tau}_m)},$$

$\widehat{\mathbb{V}\text{ar}}(\hat{\tau}_m)$ $\hat{\tau}_m$. $\mathcal{H}_0$, D_{conf} *Tracy–Widom .*

"Missing""". , $\rho(\Sigma_{\text{dis}})$ **Honest Strike:** ——

5

5.1

$K \geq 2$ $Z_1, \ldots, Z_K$, Z_k :

(IV1) **Relevance:** $\mathbb{C}\text{ov}(Z_k, T) \neq 0$;

(IV2) **Exclusion:** $Z_k \perp\!\!\!\perp Y(t) \mid T, X$;

(IV3) **Ignorability:** $Z_k \perp\!\!\!\perp U$.

$\hat{\tau}_{\text{IV}}^{(k)}$ Z_k . $\hat{\tau}_{\text{IV}}^{\text{full}}$ K , $\hat{\tau}_{\text{IV}}^{(-k)}$ $\{Z_1, \ldots, Z_K\} \setminus \{Z_k\}$.

5.2

T 9 (IV)**.** *k :*

$$Q_k = \frac{\hat{\tau}_{IV}^{full} - \hat{\tau}_{IV}^{(-k)}}{\sqrt{\widehat{\mathbb{V}\mathrm{ar}}(\hat{\tau}_{IV}^{full} - \hat{\tau}_{IV}^{(-k)})}}.$$

Z_k (IV1)–(IV3) Z_k ,

$$Q_k \xrightarrow{d} \mathcal{N}(0,1) \quad as\ n \to \infty.$$

Z_k (IV2), $Z_k \to Y$, $\gamma_k \neq 0$,

$$Q_k \xrightarrow{d} \mathcal{N}\left(\frac{\gamma_k}{\sqrt{\widehat{\mathbb{V}\mathrm{ar}}(\cdots)}}, 1\right),$$

γ_k ScaleGrowth.

Proof Sketch. IV , $\hat{\tau}_{\mathrm{IV}}^{\mathrm{full}}$ $\hat{\tau}_{\mathrm{IV}}^{(-k)}$ τ^* . Z_k , $\hat{\tau}_{\mathrm{IV}}^{\mathrm{full}}$ $\hat{\tau}_{\mathrm{IV}}^{(-k)}$, Z_k . Delta . □ □

C 10.

$$Q_{\max} = \max_{k=1,\ldots,K} |Q_k|.$$

"", $Q_{\max}$ K $Q_{\max}$, .

5.3 Disagreement

M . m Verdict $\boldsymbol{v}_m \in \{0,1\}^K$, $v_{m,k} = 1$ Z_k . **Disagreement**:

$$D_{\mathrm{IV}} = \frac{1}{M(M-1)} \sum_{i<j} \|\boldsymbol{v}_i - \boldsymbol{v}_j\|_1.$$

D_{IV} Disagreement. D_{IV} Q_k k; D_{IV} Q_k .

Honest Strike / Honest Critique:
Honest Strike: IV $F < 10$, , Q_k **Honest Strike:** . Q_k **Consist**

6 do-calculus

6.1 Pearl

G DAG, X, Y, Z, W . do-calculus (2):

- **1/:** $P(y \mid \mathrm{do}(x), z, w) = P(y \mid \mathrm{do}(x), w)$ $G_{\overline{X}}$ $Y \perp\!\!\!\perp Z \mid X, W$.
- **2:** $P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) = P(y \mid \mathrm{do}(x), z, w)$ $G_{\overline{X}\underline{Z}}$ $Y \perp\!\!\!\perp Z \mid X, W$.
- **3/:** $P(y \mid \mathrm{do}(x), \mathrm{do}(z), w) = P(y \mid \mathrm{do}(x), w)$ $G_{\overline{XZ(W)}}$ $Y \perp\!\!\!\perp Z \mid X, W$.

6.2

T 11 (do-calculus–SCX)**.** $\mathcal{E} = \{\mathcal{M}_1, \ldots, \mathcal{M}_M\}$ M , $\mathcal{Q}$.

(1) ***1*** i, j $\mathcal{Z}_i \neq \mathcal{Z}_j$ *awaiting* $\hat{\tau}_i \approx \hat{\tau}_j$, *1* $G_{\overline{T}}$ —— $\mathcal{Z}_i \triangle \mathcal{Z}_j$

(2) ***2*** $G_{\overline{X}\underline{Z}}$ —— . *scx_causal Theorem 1*

(3) ***3*** Z *do-Disagreement, SCX* D_{IV} *Disagreement* $G_{\overline{XZ(W)}}$.

: ***SCX***

Proof Sketch. **(1)** (⇒): do r ⇒ **(2)** (⇐): Disagreement SCX ⇒ .Disagreement, Core Pearl Bareinboim ——do-calculus (2; 3)—— scx_causal . □ □

C 12 (Disagreement do)**.** *SCX* M *Disagreement* $\nexists$ *Consistency* $n \to \infty$ *, do DAG* ——

> **Honest Strike / Honest Critique:**
>
> **Honest Strike:** . scx_causal 3-SCM Core: **awaitingDisagreement**. **Honest Strike:** do-calculus (4). DAG ; , c-awaiting. **Honest Strike:** 11 Proof""

7

7.1 M

$M \to \infty$, $\mathscr{H}$ Π, :

$$\bar{\tau}_\infty = \int_{\mathcal{M} \in \mathscr{H}} \tau(\mathcal{M})\, d\Pi(\mathcal{M}),$$

ATE Π-. τ^* , Π , $\bar{\tau}_\infty$. SCX Causal Inference (5).

7.2 Compactness

4 : M , $M-1$ $\to$ max, 1 .Hoeffding —— $M-1$ Compactness $R \in \mathbb{R}^{M\times M}$, $R_{ij} = \mathrm{Corr}(\hat{\tau}_i, \hat{\tau}_j)$.

7.3

1 **:** C_M, ;

2 **:** D_{conf}, ;

3 **:** IV , $\{Q_k\}_{k=1}^K$;

4 **do** Disagreement, 11

5 **Verdict:** , ATE

7.4

(1) **:** .

(2) **:** M

(3) **:** $p \gg n$,

(4) **:** . , SCX g-

(5) **:**

8 Conclusion

Causal SCX——Causal Inference, scx_causal 3-SCM M .Proof do-calculus SCX . Core: , Causal Inference—— M , .———— Causal SCX , M , ——**Consistency** δ_M. Conclusion.: , Causal Inference""

References

[1] SCX. *scx_causal: 3-SCM Non-Identifiability and Multi-Expert Causal Audit.* 2026. (Theorem 1: 3-SCM)

[2] J. Pearl. *Causality: Models, Reasoning, and Inference.* 2nd ed. Cambridge University Press, 2009.

[3] E. Bareinboim and J. Pearl. Causal inference by surrogate experiments: z-identifiability. *UAI*, 2012.

[4] Y. Huang and M. Valtorta. Pearl's calculus of intervention is complete. *UAI*, 2006.

[5] F. Li, P. Ding, and F. Mealli. Bayesian causal inference: a critical review. *Statistical Science*, 2018.

[6] W. Hoeffding. Probability inequalities for sums of bounded random variables. *JASA*, 58(301):13–30, 1963.

[7] P. W. Holland. Statistics and causal inference. *JASA*, 81(396):945–960, 1986.

[8] G. W. Imbens and D. B. Rubin. *Causal Inference for Statistics, Social, and Biomedical Sciences.* Cambridge University Press, 2015.

[9] M. A. Hernán and J. M. Robins. *Causal Inference: What If.* Chapman & Hall/CRC, 2020.

[10] J. H. Stock and M. Yogo. Testing for weak instruments in linear IV regression. In *Identification and Inference for Econometric Models*, Cambridge University Press, 2005.

A 7 Proof

Proof. **:** $\mathcal{Z}_m \supseteq \mathcal{Z}^*_{\min}$, Pearl, 2009, Theorem 3.3.2, $\mathcal{Z}_m$: $\hat{\tau}_m \xrightarrow{p} \tau^*$ as $n \to \infty$. i, j: $|\hat{\tau}_i - \hat{\tau}_j| \xrightarrow{p} 0$. Σ_{dis} , , $\rho(\Sigma_{\text{dis}}) \xrightarrow{p} 0$. , $|\mathcal{Z}_m| \geq |\mathcal{Z}^*_{\min}|$, $G_m \xrightarrow{p} 0$. **:** $\mathcal{Z}_m \not\supseteq \mathcal{Z}^*_{\min}$, $T \leftarrow U \rightarrow Y$ $\mathcal{Z}_m$. Imbens & Rubin, 2015, Ch.12:

$$\hat{\tau}_m \xrightarrow{p} \tau^* + b_m, \quad b_m \neq 0,$$

b_m ScaleMissing. $S_{\text{conf}} = \{m : \mathcal{Z}_m \not\supseteq \mathcal{Z}^*_{\min}\}$. $i \in S_{\text{conf}}, j \notin S_{\text{conf}}$: $|\hat{\tau}_i - \hat{\tau}_j| \xrightarrow{p} |b_i| > 0$. Gershgorin , $|S_{\text{conf}}| \geq 1$ $\rho(\Sigma_{\text{dis}})$ $\min_{i \in S_{\text{conf}}} |b_i|$. **:**.Σ_{dis} . $\mathcal{H}_0$, $O_p(n^{-1/2})$, Random MatrixTracy–Widom, 1996, F_β $\beta = 1$. D_{conf} $\mathcal{H}_0$ Tracy–Widom . □ □

B 9 Delta

Proof. $T = \alpha_0 + \boldsymbol{Z}'\boldsymbol{\alpha}_1 + \boldsymbol{X}'\boldsymbol{\alpha}_2 + \varepsilon$. $Y = \beta_0 + \hat{T}\beta_1 + \boldsymbol{X}'\boldsymbol{\beta}_2 + \eta$, $\hat{\tau}_{\text{IV}} = \hat{\beta}_1$ ATE IV . , $\sqrt{n}(\hat{\tau}_{\text{IV}} - \tau^*) \xrightarrow{d} \mathcal{N}(0, \sigma^2_{\text{IV}})$. $\hat{\tau}^{\text{full}}_{\text{IV}}$ $\hat{\tau}^{(-k)}_{\text{IV}}$. $\hat{\tau}^{\text{full}}_{\text{IV}} - \hat{\tau}^{(-k)}_{\text{IV}}$ influence function:

$$\psi_k(O) = \psi^{\text{full}}(O) - \psi^{(-k)}(O),$$

$\psi^{(\cdot)}$. , $\mathbb{E}[\psi_k(O)] = 0$, $Q_k \xrightarrow{d} \mathcal{N}(0,1)$. Z_k , $\mathbb{E}[\psi^{\text{full}}(O)] = \gamma_k \cdot h()$, $\mathbb{E}[\psi^{(-k)}(O)] = 0$, $\lambda_k(n) = O(\sqrt{n}\gamma_k)$. □ □